Grégoire Szymanski

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Education

PhD in Mathematics (supervised by M. Rosenbaum and M. Hoffmann) — Dec. 2024
École Polytechnique, CMAP — *Palaiseau, France*

Diplôme de l'ENS in Mathematics — Sept. 2021
École Normale Supérieure — *Paris, France*

Master El Karoui – Master of Science in Quantitative Finance — Sept. 2020
Sorbonne Universités — *Paris, France*

Bachelor's degree in Mathematics and Computer Science — Sept. 2018
École Normale Supérieure — *Paris, France*

Academic Positions

Postdoctoral researcher (supervised by M. Podolskij) — Sept. 2024 – Sept. 2027
University of Luxembourg, Department of Mathematics — *Esch-sur-Alzette, Luxembourg*

PhD student (supervised by M. Rosenbaum and M. Hoffmann) — Sept. 2021 – Sept. 2024
École Polytechnique, CMAP — *Palaiseau, France*

Undergraduate Research Assistant — Feb. 2021 – Sept. 2021
École Polytechnique, CMAP — *Palaiseau, France*

Research Internship — Feb. 2019 – June 2019
University of Helsinki — *Helsinki, Finland*

Research Internship — June 2017 – July 2017
Paris Diderot — *Paris, France*

Publications and Preprints

[13] Carsten H. Chong, Marc Hoffmann, Mathieu Rosenbaum and Grégoire Szymanski (2026): Intraday Volatility Dynamics. **Submitted**.

[12] Joseph Leclère, Youssef Ouazzani Chahdi, Mathieu Rosenbaum and Grégoire Szymanski (2026): Exact conditional simulation of Point processes: Application to pathwise market impact estimation. **Submitted**.

[11] Johannes Muhle-Karbe, Youssef Ouazzani Chahdi, Mathieu Rosenbaum and Grégoire Szymanski (2026): A unified theory of order flow, market impact, and volatility. **Submitted**.

[10] Florian Bourgey, Patrick Noble, Ingi Petursson, Mathieu Rosenbaum and Grégoire Szymanski (2026): The Quadratic Rough Heston+ Model for Short-Dated Options. **To appear in Risk**.

[9] Nathan De Carvalho, Youssef Ouazzani Chahdi and Grégoire Szymanski (2026): Trading with market resistance and concave price impact. **Submitted**.

[8] Grégoire Szymanski and Wei Xu (2026): Mean-Field Limits for Nearly Unstable Hawkes Processes. **Electronic Journal of Probability**.

[7] Youssef Ouazzani Chahdi, Mathieu Rosenbaum and Grégoire Szymanski (2026): Passive Market Impact: A Point Process Approach. **To appear in Finance and Stochastics**.

[6] Raphael Maillet and Grégoire Szymanski (2024): Estimation of the invariant measure of a multidimensional diffusion from noisy observations. **Submitted**.

[5] Grégoire Szymanski and Tetsuya Takabatake (2023): Asymptotic Efficiency for Fractional Brownian Motion with general noise. **Submitted**.

[4] Bruno Durin, Mathieu Rosenbaum and Grégoire Szymanski (2026): The two square root laws of market impact and the role of sophisticated market participants. **The Annals of Applied Probability**.

[3] Carsten H. Chong, Marc Hoffmann, Yanghui Liu, Mathieu Rosenbaum and Grégoire Szymanski (2024): Statistical inference for rough volatility: Central limit theorems. **The Annals of Applied Probability**.

[2] Carsten H. Chong, Marc Hoffmann, Yanghui Liu, Mathieu Rosenbaum and Grégoire Szymanski (2024): Statistical inference for rough volatility: Minimax Theory. **The Annals of Statistics**.

[1] Grégoire Szymanski (2024): Optimal estimation of the rough Hurst parameter in additive noise. **Stochastic Processes and their Applications**.

Awards and Grants

Susquehanna-BFS Junior Scholar Award for Most Outstanding Paper 2026
13th World Congress of the Bachelier Finance Society — for "A unified theory of order flow, market impact, and volatility"

PGSM Master Scholarship 2018 – 2020
Fondation Sciences Mathématiques de Paris

Research Visits

Beijing Institute of Technology Sept. 2023
*Invited by **Wei Xu*** *Liangxiang, China*

Hong Kong University of Science and Technology Sept. 2023
*Invited by **Carsten Chong*** *Hong Kong, China*

Hiroshima University Aug – Sept. 2023
*Invited by **Tetsuya Takabatake*** *Hiroshima, Japan*

Academic Service

Co-Organizer of the AMART Seminar - Applied Mathematics and Related Topics at the University of Luxembourg 2025 – 2026
with Lorenzo Cristofaro and Francesca Pistolato *Esch-sur-Alzette, Luxembourg*

Co-Organizer of the FinML Workshop - Mathematical Insights from Markets, Control, and Learning 2024
with Lorenzo Croissant and Antonio Ocello *Aussois, France*

Reviewer Since 2022
The Annals of Applied Probability, Quantitative Finance, SPA, Electronic Journal of Statistics, Finance and Stochastics, Decisions in Economics and Finance, Bernoulli, and SIAM Journal on Financial Mathematics

Teaching Experience

Continuous Time Models in Mathematical Finance (Master of Mathematics) Since 2025
Exercises, University of Luxembourg

Bayesian Statistics (Master of Mathematics) Since 2025
Exercises, University of Luxembourg

Logiciels mathématiques (Bachelor of Mathematics) 2025
Lecture and Exercises, University of Luxembourg

Probabilistic Models in Finance (Master of Mathematics) Since 2024
Lecture and Exercises, University of Luxembourg

How to Write Mathematics (Bachelor 1st year) 2021 – 2022
Lecture and Exercises, École Polytechnique

Processus stochastiques (M2 Mathématiques pour les Sciences du Vivant) 2021 – 2022
Exercises, École Polytechnique

Mathematics (High School, TALENS section) 2018 – 2019
Lecture and Exercises, ENS

Student Supervision and Mentoring

Master's students
Ilya Ilchenko, Attilio Pitelli, Anubhav Roy, Gaurav Yengera, Juan David Jaramillo Bobadilla, Catherine Monoue Konga

Bachelor's students
Karel Moryoussef, Adan Fhima, Catia Alves Pires

Professional Experience

Jump Trading Since 2025
ILB Chair "Artificial Intelligence and Quantitative Methods for Finance" *Paris, France*

Melanion Capital 2025
Consultant *Paris, France*

Welwing Capital 2022 – 2025
Consultant *Paris, France*

BNP Paribas June 2020 – Dec. 2020
Quantitative analyst intern *London, UK*

EDF Sept. 2019 – Dec. 2019
Partnership with ENS *Paris, France*

Seminars and Conference Presentations

Jump Trading Internal Seminar, Jump Trading, Chicago, IL, USA Oct. 6, 2026

Probability/Math Finance Seminar, Carnegie Mellon University, Pittsburgh, PA, USA Sept. 21, 2026

50th AMASES conference, Università La Sapienza, Roma, Italy Sept. 3 – Sept. 5, 2026

35th European Meeting of Statisticians, Lugano, Switzerland Aug. 24 – Aug. 28, 2026

Vienna-Copenhagen Conference on Financial Econometrics, University of Vienna, Vienna, Austria Aug. 13 – Aug. 15, 2026

25th European Young Statisticians Meeting, Vilnius, Lithuania July 7 – July 10, 2026

13th World Congress of the Bachelier Finance Society, University of Bologna, Bologna, Italy June 29 – July 3, 2026

DynStoch 2026, Chalmers University of Technology and the University of Gothenburg, Gothenburg, Sweden June 15 – June 17, 2026

30th Nordic Conference in Mathematical Statistics, University of Helsinki, Helsinki, Finland	June 1 – June 4, 2026
Financial Econometrics Conference, Toulouse School of Economics, Toulouse, France	May 22 – May 23, 2026
13th Days of Econometrics for Finance, Mohammed V University in Rabat, Rabat, Morocco	Apr. 28 – Apr. 30, 2026
Statistical Inference for Particle Systems, Institut Henri Poincaré and Université Évry Paris-Saclay, Paris and Évry, France	Apr. 7 – Apr. 8, 2026
XXVII Workshop on Quantitative Finance, University of Bergamo, Bergamo, Italy	Mar. 30 – Apr. 1, 2026
Séminaire sur les processus de Hawkes du LPSM, Sorbonne Université, Paris, France	Mar. 16, 2026
KCL Mathematical Finance Seminar, King's College London, London, UK	Mar. 2, 2026
2nd Dolomites Winter School: Mean-Field Systems in Finance, Neurosciences, and AI, Centro Congressi - Folgarida and Madonna di Campiglio, Folgarida, Italy	Jan. 11 – Jan. 16, 2026
IMS International Conference on Statistics and Data Science 2025, Sevilla, Spain	Dec. 15 – Dec. 18, 2025
CFE-CMStatistics 2025, Birkbeck, University of London, London, UK	Dec. 13 – Dec. 15, 2025
2025 European Winter Meeting of the Econometric Society, University of Cyprus, Nicosia, Cyprus	Dec. 7 – Dec. 9, 2025
Applied Mathematics and Related Topics Seminar, University of Luxembourg, Esch-sur-Alzette, Luxembourg	Oct. 21, 2025
Young Researchers in Stochastic Analysis and Stochastic Geometric Analysis, Bernoulli Center, EPFL, Lausanne, Switzerland	Sept. 23 – Sept. 26, 2025
Mathematical Finance Seminar, Columbia University, New York, NY, USA	Sept. 4, 2025
SIAM Conference on Financial Mathematics and Engineering, Miami, FL, USA	July 15 – July 18, 2025
17th Annual SoFiE Conference, ESSEC Campus of Cergy, Cergy, France	June 9 – June 12, 2025
Luxembourg Workshop in Stochastic Analysis, University of Luxembourg, Esch-sur-Alzette, Luxembourg	Feb. 12, 2025
Prob&Stat Seminar, University of Luxembourg, Esch-sur-Alzette, Luxembourg	Oct. 17, 2024
FinML Workshop, Centre Paul Langevin, Aussois, France	Sept. 23 – Sept. 27, 2024
Third-year PhD Students' Day, Université Paris Dauphine, Paris, France	Mar. 26, 2024
SMSA 2024, TU Delft, Delft, Netherlands	Mar. 13 – Mar. 15, 2024
Matinée recherche CFM, Capital Fund Management, Paris, France	Mar. 12, 2024

EFFI France-Japan seminar, Le Mans Université, Le Mans, France	Dec. 5, 2023
Séminaire des doctorants du CEREMADE, Université Paris Dauphine, Paris, France	Nov. 23, 2023
Séminaire des doctorants du CMAP, École Polytechnique, Palaiseau, France	Nov. 22, 2023
Seminar, Beijing Institute of Technology, Beijing, China	Sept. 21, 2023
ICIAM 2023, Waseda University, Tokyo, Japan	Aug. 20 – Aug. 25, 2023
Summer School on Big Data and Finance, Centre Paul Langevin, Aussois, France	June 12 – June 16, 2023
Workshop on modeling with Hawkes processes: Theory, Statistics and Applications, Institute of Mathematics of the Polish Academy of Sciences, Warsaw, Poland	May 29 – June 2, 2023
Groupe de Travail du CERMICS, École des Ponts, Champs-sur-Marne, France	May 15, 2023
Séminaire des doctorants du CMAP, École Polytechnique, Palaiseau, France	Feb. 22, 2023
Groupe de travail des thésards du LPSM, Jussieu, Paris, France	Nov. 7, 2022
Séminaire des jeunes chercheurs du Ceremade, Université Paris Dauphine, Paris, France	Oct. 13, 2022
StatMathAppli 2022, La Villa Clythia, Fréjus, France	Aug. 29 – Sept. 2, 2022
DynStoch 2022, Institut Henri Poincaré, Paris, France	June 29 – July 1, 2022
CEREMADE Winter School, Domaine de la Tour, Saint-Pierre-Canivet, France	Feb. 27 – Mar. 4, 2022